Wine Quality Assessment Using Data Mining Techniques

NameCSC 6740 - Data Mining

***Abstract*—This paper investigates wine quality classification using data mining techniques on the UCI Wine Quality dataset. The original quality scores were transformed into a three-class classification problem (low, medium, high) to address class imbalance while preserving meaningful distinctions in quality. Multiple machine learning models were evaluated, including Logistic Regression, Support Vector Machines, Random Forest, Gradient Boosting, and XGBoost. Data preprocessing included feature scaling, stratified sampling, and Principal Component Analysis. Feature analysis further indicates that alcohol, sulphates, density, and acidity-related variables contribute strongly to wine quality prediction.**

***Index Terms*—Data mining, wine classification**

I. Introduction

WINE quality assessment is traditionally based on sensory evaluation by human experts, but this process can be subjective and time-consuming[1]. Data mining and machine learning provide an alternative by predicting wine quality from measurable physicochemical properties. In this project, I aim to investigate whether wine quality can be predicted using these chemical attributes such as fixed acidity, volatile acidity, citric acid, residual sugar, chlorides, free sulfur dioxide, total sulfur dioxide, density, pH, sulphates, alcohol [1].

The dataset used in this project is the Wine Quality dataset from the UCI Machine Learning Repository [2]. It contains red and white vinho verde wine samples from Portugal, each represented by 11 physicochemical variables and a quality score from 0 (very bad) to 10 (excellent) [1][2]. Following the original study by Cortez et al., the red and white wine datasets are analyzed separately because they have different physicochemical profiles and taste characteristics [1].

This project builds upon the work of Cortez et al., published in Decision Support Systems (2009), titled "Modeling wine preferences by data mining from physicochemical properties" who showed that machine learning models, especially Support Vector Machines, can predict wine quality with meaningful performance. However, their work focused primarily on regression [1]. In contrast, this project reformulates the task as a multi-class classification problem and extends the comparison to additional machine learning methods, including tree based models.

A. Background and Motivation

According to Cortez et al. (2009a), wine certification involves both physicochemical testing and sensory evaluation. However, sensory analysis is subjective and influenced by human perception, which is "the least understood of the human senses" [1]. According to Cortez et al. (2009a), if machine learning models can accurately predict wine quality, then certification processes can be supported by objective tools, producers can adjust production variables (e.g., alcohol concentration) to improve quality, marketing strategies can target specific consumer preference profiles, and wine evaluation quality and speed can be improved[1].

From a technical perspective, the dataset presents useful machine learning challenges such as class imbalance, overlapping classes, and potentially nonlinear relationships among features.

B. Problem Definition

Let $D = \{(x_i, y_i)\}_{i=1}^{n}$ denote the wine dataset, where $x_i \in R^{11}$ is the vector of physicochemical features for sample i, and $y_i \in \{0, 1, \ldots, 10\}$ is the corresponding wine quality label.

Instead of predicting the original score directly, this project reformulates the task into a three-class classification problem:

- Low quality: $y \leq 4$
- Medium quality: $y \in \{5, 6\}$
- High quality: $y \geq 7$

The goal is to learn a classifier

$$f(x_i) \mapsto \hat{y}_i, \quad \hat{y}_i \in \{\text{Low}, \text{Medium}, \text{High}\}$$

that predicts the quality class of each wine sample from its physicochemical measurements.

C. System Overview

The system begins with data ingestion and preprocessing. The red and white wine datasets are loaded separately, cleaned, and transformed into the three-class target representation. Because the class distribution is imbalanced, stratified train-test splitting is used to preserve class proportions during training and testing.

After preprocessing, the feature space is standardized using `StandardScaler`[3]. Principal Component Analysis (PCA) [3] is then applied to reduce dimensionality while preserving most of the variance. The transformed feature space is used to train and evaluate several machine learning models:

- Logistic Regression: Baseline Model
- Support Vector Machine: Paper Comparison
- Random Forest: Strong Tree Model
- Gradient Boosting: Strong Tree Based Model
- XGBoost: Strong Tree Based Model

The Scikit-learn implementations were used for Logistic Regression, SVM, Random Forest, and Gradient Boosting [3], while XGBoost was implemented using the XGBoost package [4]. All models are tuned using GridSearchCV [3]. Model performance is evaluated using accuracy, balanced accuracy, macro F1-score, weighted F1-score, and confusion matrices. These metrics provide a broader view of model quality, especially under class imbalance.

PCA is only performed once during baseline modeling with Logistic Regression. Since PCA measure which features contribute to the final result, doing it once is sufficient to understand feature importance.

D. Related Context

Cortez et al. conducted a comprehensive study on wine quality modeling [1]. Their main contributions include testing multiple models for wine tasting (Multiple Regression, Neural Networks, and Support Vector Machines), simultaneous model and variable selection using sensitivity analysis, evaluation using tolerance-based accuracy, and identification of key features influencing wine quality [1].

Their findings showed that the support vector machine model outperformed neural network and multiple regression models [1]. For tolerance T = 0.5, SVM's overall accuracy for red wines is 62.4% and 64.6% for white wines. For tolerance T = 1.0, SVM's overall accuracy for red wines is 89.0% and 86.8% for white wines [1].

They also found that sulphates are among the most important predictors for both red and white wines [1]. However, there are several limitations and opportunities for extension.

The paper only compares three algorithms, does not explore tree based modeling, and focuses on a regression problem rather than a classification problem.

This project extends that work in several ways. First, it reformulates the problem as a three-class classification task rather than a regression task. Second, it evaluates additional tree-based approaches such as Random Forest, Gradient Boosting, and XGBoost.

II. Methodology

A. Proposed Algorithms and Methods

Five machine learning approaches were evaluated in this project: Logistic Regression, Support Vector Machine (SVM), Random Forest, Gradient Boosting, and XGBoost.

Logistic Regression was used as a baseline model because it is simple and interpretable. SVM was included because it performed strongly in prior work on wine quality prediction [1].

Random Forest, Gradient Boosting, and XGBoost were selected as stronger nonlinear tree based modeling methods that may better capture complex relationships in the physicochemical data.

B. Dataset Handling

The Wine Quality dataset contains two separate datasets:

- Red wine dataset: 1599 samples

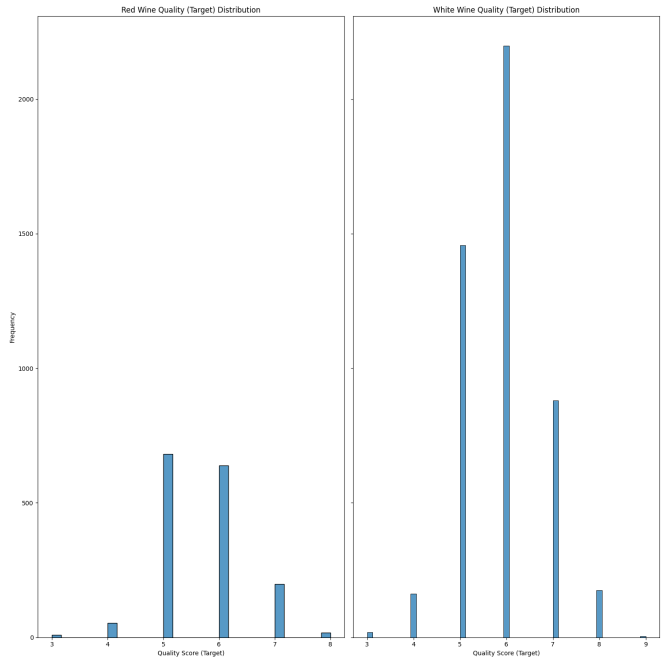

Fig. 1. Distribution of wine quality classes before transforming the original quality scores into Low, Medium, and High categories.

- White wine dataset: 4898 samples

Each sample contains the following 11 physicochemical features:

- fixed acidity
- volatile acidity
- citric acid
- residual sugar
- chlorides
- free sulfur dioxide
- total sulfur dioxide
- density
- pH
- sulphates
- alcohol

The red and white datasets were modeled separately to remain consistent with the literature and because the two wine types have distinct chemical profiles [1].

C. Target Transformation

The original quality scores were converted into three classes:

- Low: quality ≤ 4
- Medium: quality $= 5$ or 6
- High: quality ≥ 7

This data transformation was made after looking at the class score distribution in the dataset, as shown in Fig. 1. Most of the data is concentrated in the 5-6 score range in both the red and white wine datasets. Thus, there is a class imbalance where more wines are considered normal, 'medium', rather than on extreme ends (bad quality or great quality).

This transformation, shown in Fig. 2, simplifies the original 0–10 scoring system while still preserving meaningful distinc-

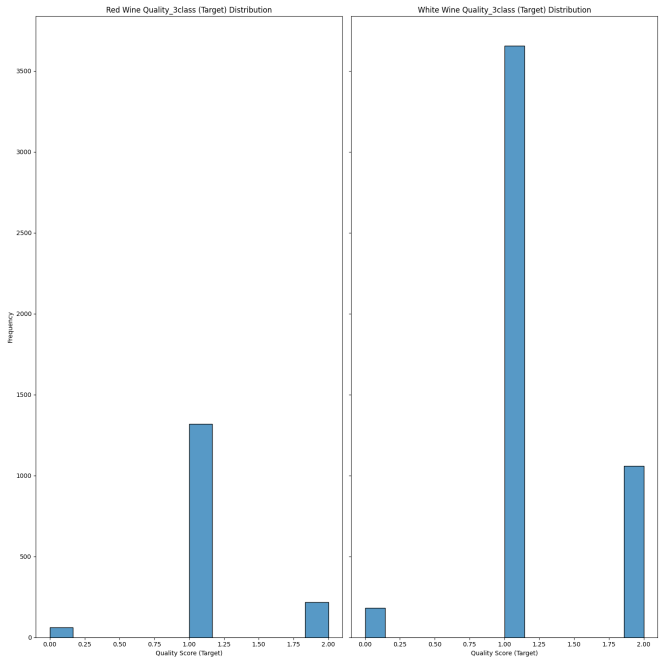

Fig. 2. Distribution of wine quality classes before transforming the original quality scores into Low, Medium, and High categories.

tions in quality. It also reduces, though does not eliminate, the class imbalance present in the original data.

D. Class Imbalance Handling

As mentioned previously, the datasets are highly imbalanced. There are many more samples in the medium-quality class. For example, see 2, in the red wine dataset, the medium class dominates the majority of the samples. This imbalance is even more severe in the white wine dataset.

To address this issue, stratified train-test splitting was used to preserve class proportions in both the training and test sets. For Logistic Regression, `class_weight='balanced'` was used to reduce bias toward the majority class.

SMOTE was also applied to reduce the effect of class imbalance by oversampling minority classes [5].

E. Feature Scaling and Dimensionality Reduction

Before model training, all input features were standardized using `StandardScaler`[3]. Since the original features are measured on different scales, standardization ensures that no single feature dominates due to its magnitude alone.

Principal Component Analysis (PCA)[3] is then applied with `n_components = 0.95`, meaning enough principal components were retained to preserve 95% of the total variance in the Logistic Regression model. Principal Component Analysis provided insight into which physicochemical variables contributed most strongly to variation in the data.

The PCA loadings for the red wine, 3 showed that features such as fixed acidity, citric acid, density, and pH strongly contribute to the principal components. This is consistent with the study as [1] found that pH was an important feature (using average SVM relative importance plots).

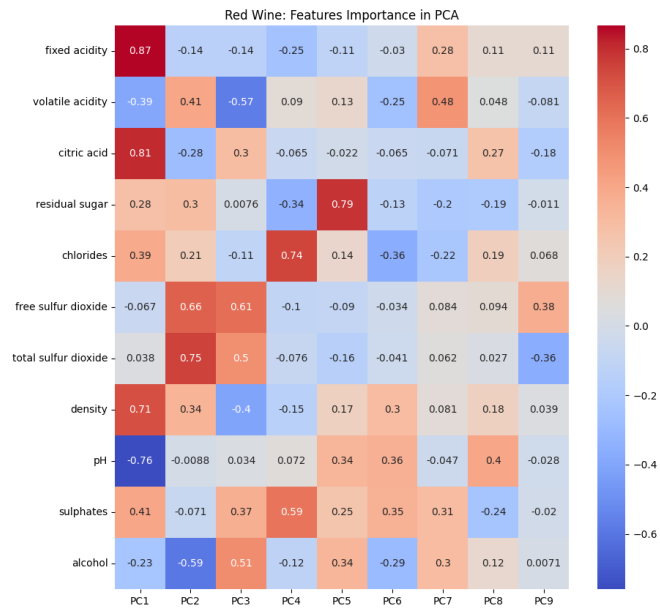

Fig. 3. PCA loading heatmap for the red wine dataset.

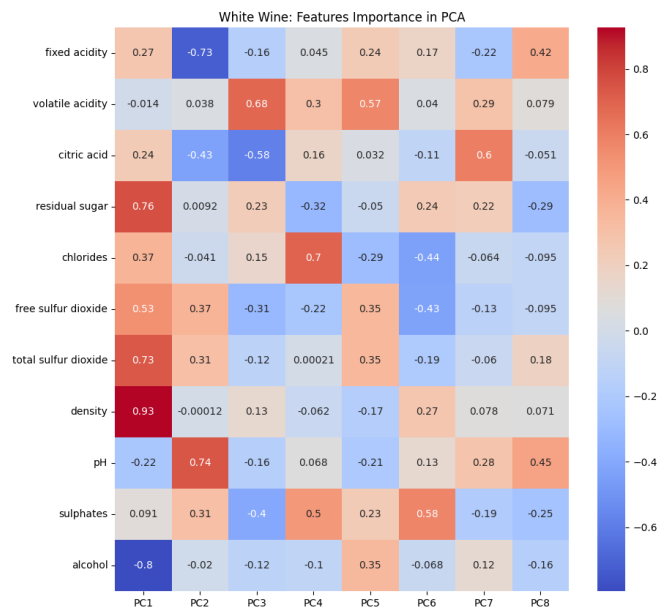

Fig. 4. PCA loading heatmap for the white wine dataset.

The PCA loadings for the white wine, 4 showed that features such as density, alcohol, total sulfur dioxide, and residual sugar strongly contribute to the principal components. This is consistent with the study as [1] found that alcohol and residual sugar were important features (using average SVM relative importance plots).

F. Mathematical Models Used

For Logistic Regression model used in this project can be defined as:

$$f_{\vec{w},b}(\vec{x}) = \frac{1}{1 + e^{-(\vec{w}\cdot\vec{x}+b)}}$$

where $\vec{w}$ is the weight vector, b is the bias term, and $\vec{x}$ is the input instance.

For Support Vector Machines, the classifier seeks a decision boundary of the form

$$f(x) = w^T x + b = 0$$

where w^T and b are used to define the hyperplane separating the two classes and x is the support vectors. Support Vector Machines can be mapped to a higher dimensional space.

Random Forest, Gradient Boosting, and XGBoost are modeling methods based on multiple decision trees.

G. Flowchart

Figure 5 illustrates the overall workflow used for wine quality classification.

H. System Architecture and Components

The project can be divided into four main components:

1) **Data Ingestion:** Load the red and white wine datasets
2) **Preprocessing:** Perform label transformation and standardization (through Pipeline)
3) **Modeling:** Model training
4) **Evaluation:** Compute performance metrics and visualizations such as confusion matrices and feature importance plots

I. Design Justifications

Several design choices were made to improve both interpretability and performance.

First, the problem was converted into a three-class classification task because adjacent quality scores are difficult to distinguish reliably, while three classes retain meaningful distinctions.

Second, stratified splitting and SMOTE [5] were used because the dataset is highly imbalanced.

Third, PCA was used to make feature importance easier to analyze.

Finally, tree based models were included because the wine quality problem may involve nonlinear relationships that a simple linear model cannot capture well.

J. Analysis

The methodology was designed to compare simple and advanced models under the same preprocessing pipeline. This makes it easier to assess how much performance gain comes from model complexity rather than from changes in preprocessing.

In particular, comparing Logistic Regression with tree-based methods helps determine whether the relationship between physicochemical properties and wine quality is primarily linear or nonlinear.

III. Evaluation

A. Experimental/Demo setup

Experiments were conducted using Python and standard machine learning libraries, including `pandas`, `NumPy`, `scikit-learn` [3], and `XGBoost` [4]. A stratified train-test split was used so that each class maintained a similar distribution in the training and testing sets.

The evaluation metrics used in this project were:

- Accuracy
- Balanced Accuracy
- Macro F1-score
- Weighted F1-score
- Classification Report
- Confusion Matrix

Accuracy provides overall correctness, while balanced accuracy accounts for uneven class sizes.

Macro F1-score finds the weighted mean for each label, and weighted F1-score was used because it changes the macro f1-score to take class imbalance into account[3].

Classification Report provides an overview of the model performance across all classes using precision, recall, f1-score, and support. Confusion matrices were used to examine which classes were most frequently confused by the models.

B. Results

The models were evaluated both before and after applying SMOTE to address class imbalance [5].

1) What is SMOTE?: SMOTE is Synthetic Minority Oversampling Technique which generates new samples of a defined class by 'interpolating between several minority class examples that lie together' [5].

This oversampling technique is used here since the number of low and high samples is so little. By synthetically creating more samples for the lower occurring classes, we provide the model with more training data so that the model becomes better at recognizing and accurately classifying them.

2) Before applying SMOTE: Before applying SMOTE, the highest accuracy achieved on the red wine dataset was with XGBoost with an accuracy of 0.842, while Gradient Boosting achieved the highest accuracy on the white wine dataset with an accuracy of 0.833. However, these high accuracy values were partly caused by the dominance of the Medium class in the dataset.

Since most samples belong to the Medium class, models could achieve high accuracy while still performing poorly on the Low and High minority classes. This can be seen as the balanced accuracy for XGBoost[4] before SMOTE on red wine dataset is 0.499 and 0.572 for Gradient Boosting.

Data Ingestion → Preprocessing → Target Transformation → Train/Test Split → Scaling → PCA (Logistic Regression) → SMOTE → Hyperparameter Tuning → Model Training → Evaluation

Fig. 5. Project workflow for wine quality classification.

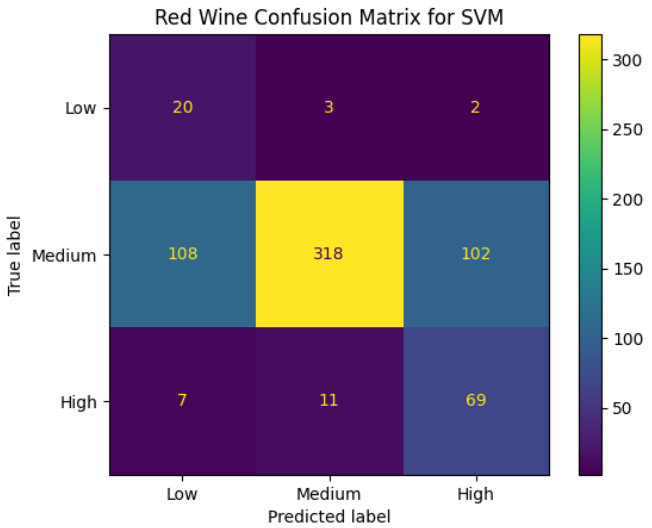

Fig. 6. Confusion matrix for the SVM model on the red wine dataset after applying SMOTE.

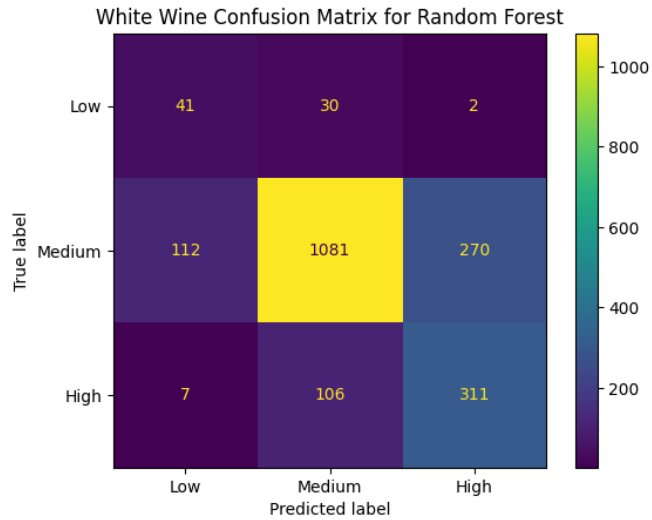

Fig. 7. Confusion matrix for the Random Forest model on the white wine dataset after applying SMOTE.

3) After applying SMOTE: After applying SMOTE, overall accuracy generally decreased, but balanced accuracy improved for most models. For the red wine dataset, SVM achieved the highest balanced accuracy after SMOTE with a balanced accuracy of 0.732. Random Forest also improved from 0.557 balanced accuracy before SMOTE to 0.670 after SMOTE. For the white wine dataset, Random Forest achieved the best balanced accuracy after SMOTE, improving from 0.591 to 0.678.

These results show that SMOTE helped the models pay more attention to the minority classes. The tradeoff is that the models became less accurate overall because they no longer favored the majority Medium class as strongly.

Therefore, accuracy alone is not the best metric for this project. Balanced accuracy and macro F1-score are more meaningful because they evaluate performance across all three wine quality classes more fairly.

Overall, Random Forest provided the best balance between accuracy and class-balanced performance after SMOTE. Although XGBoost and Gradient Boosting had higher accuracy before SMOTE, their balanced accuracy was lower, showing that they were more biased toward the majority class. The final results suggest that SMOTE improves fairness across quality categories, even though it may reduce overall accuracy.

C. Observations

The confusion matrices show that most errors occurred between adjacent quality categories, especially Low versus Medium and Medium versus High. This is expected because wine quality scores are subjective and neighboring classes may have similar chemical properties. After SMOTE, the models predicted Low and High classes more often, which improved balanced accuracy. However, this also increased misclassification of some Medium samples, causing total accuracy to decrease. The confusion matrices in Fig. 6 and Fig. 7 show that most errors occur between neighboring classes even after SMOTE.

The results demonstrate an important tradeoff between overall accuracy and class-balanced performance. Without SMOTE, the models favored the Medium class. With SMOTE, the models became better at recognizing minority classes, making balanced accuracy a more appropriate metric for this project.

D. Conclusions

This project showed that wine quality classification is affected strongly by class imbalance. Models trained without SMOTE achieved higher overall accuracy, but this performance was partly misleading because the datasets were dominated by Medium-quality wines. After applying SMOTE, balanced accuracy improved for several models, especially SVM and Random Forest, indicating better performance across Low, Medium, and High classes.

Random Forest after SMOTE was selected as the most balanced final model because it maintained strong accuracy while improving minority-class recognition. The results suggest that physicochemical properties can be used to predict wine quality categories, but distinguishing neighboring quality levels remains challenging.

IV. Learning Outcomes

A. Tasks Completed by Each Team Member

This was an individual project. All project tasks were completed independently, including literature review, problem formulation, data preprocessing, exploratory analysis, model implementation, hyperparameter tuning, result interpretation, report writing, and demo preparation.

B. Reasons for Any Incomplete Tasks or Failures

No project components were left incomplete.

C. Lessons Learned

This project provided practical experience in applying machine learning to a real-world classification problem with class imbalance. One major lesson learned was that accuracy alone is not sufficient for evaluation when classes are unevenly distributed. Another lesson was that preprocessing decisions, such as label transformation, and feature scaling can strongly affect model performance.

D. Teamwork Experience

Since this was an individual project, there was no team collaboration component. However, completing the project independently strengthened my skills in experimental design, debugging, project planning, and technical communication through report writing.

V. Group Member Evaluation

This was an individual project, thus there are no group members for evaluation.

VI. Demo Video Access Link

The demo video can be accessed at the following link:

`[https://youtu.be/lnbEtSB-15k]`

References

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